

ANKUR DIXIT

112, Hollister Dr, Ithaca, NY, 14850, USA
Email id: ad933@cornell.edu
Phone Number: +1-(607)-262-3543

Academic qualifications

2016. 01 – 2022. 04 **PhD in Atmospheric Sciences**, Indian Institute of Technology Delhi, India

Research interest/experience

1. Working with mesoscale model like Weather Research and Forecasting (WRF), WRF-Hydro to study hydro-climate change and variability.
2. Expertise in C, Python, and MATLAB programming languages, GIS software and state-of-art parameter optimization techniques.

Current position and past employment history

2023. 01 - Current **Postdoctoral Associate**, EAS, Cornell University, Ithaca, NY, USA

2021. 11 – 2023. 01 **Senior Associate**, PwC India, India.

2014. 08 – 2015. 12 **JRF** Water Resources Department, Indian Institute of Remote Sensing, ISRO, Dehradun, India

Publications (selected)

1. Dixit, A., Musselman, K., Addor, N., Carr, S., Gutmann, E., ... & Lehner, F. Future mountain snowpacks are more resilient than climate model suggest. (Under preparation)
2. Dixit, A., Najibi, N., & Lehner, F. Understanding the Causality of the 2024 Contrasting drivers of the 2024 heatwave over North-Central India. (Under preparation)
3. **Dixit, A.**, Rahimi, S., Huang, L., Musselman, K. N., Vano, J. A., Addor, N., & Lehner, F. (2026). High-resolution models better simulate historical snowpack declines in the Upper Colorado River Basin. *Environmental Research Letters*.
4. **Dixit, A.**, Sahany, S., Mishra, S. K., & Lehner, F. (2025). Optimizing WRF-Hydro Calibration in the Himalayan Basin: Precipitation Influence and Parameter Sensitivity Analysis. *Journal of Hydrometeorology*.
5. **Dixit, A.**, Sahany, S., & Mishra, S. K. (2023). Modeling the climate change impact on hydroclimate fluxes over the Beas basin using a high-resolution glacier-atmosphere-hydrology coupled setup. *Journal of Hydrology*, 627, 130219.
6. **Dixit, A.**, Sahany, S., Mishra, S. K., & Mesquita, M. D. (2023). Seasonal dependent suitability of physical parameterizations to simulate precipitation over the Himalayan headwater. *Scientific Reports*, 13(1), 4756.
7. **Dixit, A.**, Sahany, S., Rajagopalan, B., & Choubey, S. (2022). Role of changing land use and land cover (LULC) on the 2018 megafloods over Kerala, India. *Climate Research*, 89, 1-14.
8. **Dixit, A.**, Sahany, S., & Kulkarni, A. V. (2021). Glacial changes over the Himalayan Beas basin under global warming. *Journal of Environmental Management*, 295, 113101.

Presentations (selected)

1. **Dixit, A.**, Rahimi, S., Haung, L., Addor, N., Vano, J. A., ... & Lehner, F. (2024, December). Model resolution sensitivity of time of emergence for snow metrics in the Upper Colorado River Basin. In AGU Fall Meeting Abstracts.
2. **Dixit, A.**, Najibi, N., & Lehner, F. (2024, December). Understanding the Causality of the 2024 Long-Lasting Heatwave Over India: The Role of Anthropogenic Climate Change and Natural Climate Variability. In AGU Fall Meeting Abstracts.
3. **Dixit, A.**, Rahimi, S., Haung, L., Collins, E., Addor, N., Vano, J. A., ... & Lehner, F. (2023, December). Model resolution dependence of snow-water resilience to climate change in the Upper Colorado River Basin. In AGU Fall Meeting Abstracts (Vol. 2023, No. 6, pp. A23U-06).

Reports

DOE (Department of Energy). (2024). Understanding Decision-Relevant Regional Climate Data Products: Workshop Report. DOE/SC-2021.

- (i) Context of the Workshop
- (ii) The State of Decision-Relevant Climate Data Production
- (iii) The State of Knowledge on Users' Climate Information Needs
- (iv) The State of Benchmarking and Evaluation for Regional Climate Data Sets and Projections

Contribution and Services

1. Con-convended a session "NH31F - Multihazard Early Warning System in the Himalayan Region: From Early Warning to Early Action Poster" at AGU 2024.
2. Served as reviewer for multiple reputed journals.

Contact:

+1-607-262-3543

ad933@cornell.edu / ankur.dixit@hotmail.com

The logo consists of a stylized lowercase 'g' with a small square above it, followed by a large 'R' with a superscript 'G'.